Cosmological Vacuum Selection and the Maximal Chronometric-Shear Ridge

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Archive reconstruction from the surviving project ledger. Historical sandbox binaries were ephemeral; see RECOVERY_PROVENANCE.md.

27. Homogeneous evolution

The cosmological mode obeys

$$\ddot{x} + 3H\dot{x} + \frac{1}{f_a^2}\frac{\partial V_{\rm eff}(x,t)}{\partial x} = 0,$$

with

$$V_{\rm eff} = V_0 + V_{\Psi,T} + V_{{\rm QCD},T} + V_b.$$

The protected vacuum is

$$V_0(x) = \frac{m_a^2 f_a^2}{N^2}(1 + \cos Nx),$$

with minima $x_v = (2j+1)\pi/N$.

28. Reheating phasor

Let sector temperatures be $T_k = \xi_k T_0$. The leading field-dependent high-temperature threshold potential is governed by

$$W_2 = \sum_{k=0}^{N-1} \xi_k^2 e^{2\pi ik/N},$$

and selects

$$x_T = -\arg W_2.$$

The successful sparse population is

$$\xi_0 = 1, \qquad \xi_5 = 0.25, \qquad \xi_{1,2,3,4} \ll 1.$$

For $N = 6$,

$$x_T = 0.0524383.$$

The bias scales as ξ^2, while dark radiation scales as ξ^4. One adjacent complete Standard Model copy at $\xi = 0.25$ gives the project value

$$\Delta N_{\rm eff} = 0.0289,$$

whereas populating all five hidden replicas at that temperature would be excessive under the cited cosmological bound [@GoldsteinHill2026].

29. QCD crossover as a second lens

Write the QCD pressure as

$$p(T, \Lambda) = T^4 P(T/\Lambda).$$

Dimensional analysis gives

$$\frac{\partial F}{\partial \ln \Lambda} = \rho - 3p \equiv \Theta(T).$$

Since

$$\Lambda_k(x) = \Lambda_0 \left[1 - \epsilon \cos\left(x + \frac{2\pi k}{N}\right)\right]^{2/27},$$

the leading QCD term is

$$\boxed{V_{{\rm QCD},k} = \frac{2}{27}\Theta(T_k) \ln\left[1 - \epsilon \cos\left(x + \frac{2\pi k}{N}\right)\right].}$$

The lattice-QCD trace anomaly produces a second focusing epoch near $T \simeq 0.179$ GeV in the strong benchmark [@Borsanyi2016].

30. Baryon locking and late release

Visible baryons contribute

$$V_b(x, a) = \rho_b(a)(1 - \epsilon \cos x)^{2/27}.$$

Near the even-N thermal point, baryons supply positive curvature. The vacuum potential eventually becomes tachyonic there and releases the field into an adjacent minimum. In the preferred benchmark, this occurs around

$$z_{\rm inst} \simeq 450.$$

The chronology is

$$\boxed{\text{heavy-threshold focusing} \to \text{QCD refocusing} \to \text{baryon locking} \to \text{late vacuum release.}}$$

31. Strong-attractor benchmark and ridge

The original Planck-scale benchmark was cosmologically unsatisfactory: thermal focusing was negligible and generic misalignment overclosed the Universe. The successful strong-attractor point is

$$\boxed{\begin{aligned} N &= 6, \\ f_a &= 2.435 \times 10^{10}\,\mathrm{GeV}, \\ \epsilon &= 2.70 \times 10^{-13}, \\ M &= 1.002 \times 10^{6}\,\mathrm{GeV}, \\ T_R &= 1.002 \times 10^{8}\,\mathrm{GeV}, \\ m_a &= 7.5 \times 10^{-29}\,\mathrm{eV}, \\ d_g &= 10^{-6}. \end{aligned}}$$

The focusing measures are

$$\eta_{\Psi,\mathrm{max}} = 612, \qquad \eta_{\mathrm{QCD,max}} = 759.$$

Twelve tested homogeneous initial phases converged to

$$x_v = +\frac{\pi}{6}.$$

This is a numerical existence result, not a global theorem. A moderate point with $\eta_{\Psi,\mathrm{max}} \approx 30.6$ did not select a unique branch. Merely having $m_T/H > 1$ is insufficient.

The viable $N = 6$ ridge at representative m_a and the stated cuts is approximately

$$1.6 \times 10^{9}\,\mathrm{GeV} \lesssim f_a \lesssim 9.4 \times 10^{12}\,\mathrm{GeV},$$

$$2.6 \times 10^{3}\,\mathrm{GeV} \lesssim M \lesssim 1.6 \times 10^{7}\,\mathrm{GeV}.$$

Cosmology chooses a viable branch and sign; equivalence-principle tests cap the magnitude of shear.

32. Observable scale and domain conditions

For annual solar-potential modulation $\Delta\Phi_\odot = 3.29 \times 10^{-10}$,

$$\left|\Delta \ln \frac{\nu_A}{\nu_B}\right|_{\mathrm{yr}} = 2|K_{AB}| d_g^2 \Delta\Phi_\odot.$$

At $d_g = 10^{-6}$,

$$\boxed{\left|\Delta \ln \frac{\nu_A}{\nu_B}\right|_{\mathrm{yr}} = 6.58 \times 10^{-22} |K_{AB}|.}$$

Ordinary atomic sensitivities are far below current reach. Provisional thorium strong-sector enhancements of order 10^4 would raise the illustrative signal to 6.58×10^{-18}, but the nuclear coefficients remain model dependent [@Arakawa2026; @Thorium2026].

The six vacua require either pre-inflation symmetry breaking with no restoration or a gauged/discrete completion. A sufficient homogeneity condition is

$$\frac{H_{\rm inf}}{2\pi f_a} \ll x_T.$$

For the benchmark,

$$H_{\rm inf} \ll 8.0 \times 10^9\,\mathrm{GeV}.$$

The compact field is not the dark matter on the optimised ridge; its abundance is negligible.